Lab 2 Report

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1. Problem and Assumptions

In this lab I wrote PEAS for a delivery robot and a student support agent, classified their environments, and added memory to a room heating and cooling agent so it does not repeat the same command.

The robot works on a university campus and the support agent works for one university. These are my own assumptions. The room agent has modes HEAT, COOL, OFF and ECO (empty room). The limits 18 and 26 are synthetic lab assumptions. All data is made up.

2. PEAS Specifications

Delivery robot

Performance measure. On time deliveries, number of collisions, battery used per delivery.

Environment. Campus footpaths and crossings, pedestrians and cyclists, building doors and lifts, rain and night.

Actuators. Wheels and brakes, parcel compartment lock, horn and lights.

Sensors. Camera, lidar or ultrasonic sensors, GPS, battery sensor.

Trade-off. Driving faster gives more on time deliveries, but it is less safe near people and uses more battery. I would slow down near pedestrians even if a delivery is late.

LLM based student support agent

Performance measure. Accuracy of answers compared with the official policy, percentage of questions solved, correct handover to a human.

Environment. Chat with the student, university policy documents, student records system, ticket system.

Actuators. Text replies, creating a support ticket, booking an appointment, handing the chat to a human.

Sensors. Student messages, knowledge base search results, read only student record fields, conversation history.

Tools the agent can access. Knowledge base search, read only record lookup for the logged in student, ticket creation and appointment booking. It cannot change grades or fees and cannot see other students. This is my assumption.

Trade-off. Answering fast from memory is quick but may give an old policy. Searching the documents every time is more accurate but slower. I would search for policy questions and open the student record only when needed.

3. Environment Classification

Delivery robot is partially observable and stochastic. It can see its battery level and nearby obstacles, but it cannot see behind a corner or know if a pedestrian will step in front. A brake command is predictable, but driving on a wet path is uncertain because the wheels can slip.

Student support agent is partially observable and stochastic. It can see the message text and the search results, but it cannot see the real situation of the student. A ticket tool normally returns a ticket number, but the student reaction and the wording of the LLM answer are uncertain.

A simplified simulator is different from the real environment. For example a robot on a fixed grid map with no people would be fully observable and deterministic, and so would a support agent tested with scripted questions and a fixed answer key. These labels are true only under these assumptions and may not hold in the real campus or university.

4. Room Agent with Memory

The code is in lab2_percepts.py, lab2_model.py and lab2_check.py. The old rules are kept. Below 18 is HEAT, above 26 is COOL, and 18 to 26 including both boundaries is OFF. An empty room is always ECO because occupancy has priority. The StatefulAgent stores previous_mode and returns the target mode and a send flag. The flag is true only when the target is different from previous_mode.

Comparison. Both agents choose the same target in all 11 steps. The reflex agent sent 11 commands and the stateful agent sent only 5.

Complete mode transition trace

- Step 1. Previous none, 15.0 occupied, target HEAT, command sent.
- Step 2. Previous HEAT, 15.0 occupied again, target HEAT, no command.
- Step 3. Previous HEAT, 16.5 occupied, target HEAT, no command.
- Step 4. Previous HEAT, 18.0 occupied (lower boundary), target OFF, command sent.
- Step 5. Previous OFF, 22.0 occupied, target OFF, no command.
- Step 6. Previous OFF, 26.0 occupied (upper boundary), target OFF, no command.
- Step 7. Previous OFF, 26.5 occupied, target COOL, command sent.
- Step 8. Previous COOL, 30.0 empty, target ECO, command sent.
- Step 9. Previous ECO, 30.0 empty again, target ECO, no command.
- Step 10. Previous ECO, 12.0 empty, target ECO (occupancy priority), no command.
- Step 11. Previous ECO, 12.0 occupied, target HEAT, command sent.

All 11 steps matched the expected values.

5. Failure / Edge Case

Input. Occupied room with sensor values 17.9, 18.0, 17.9, 18.0, 17.9.

Observed behaviour. The mode changes HEAT, OFF, HEAT, OFF, HEAT and a command is sent every time (5 commands).

Likely cause. The limit 18 is a hard cut-off, and the memory only removes identical commands, it does not stop the mode from flipping.

Proposed improvement. Use hysteresis, for example start heating below 18 but go to OFF only at 19.

6. Limitations

I used only 11 made up percepts and no real heater. The limits and the PEAS labels depend on my assumptions. The stateful agent also assumes that every command it sent was received.

7. Reflection and Responsible Practice

[Write your reflection here in your own words.]

- All data is made up. No real student information or passwords are uploaded.
- The support agent should protect private data and hand over serious cases to a human. The robot should choose safety before speed.
- No outside code was copied. The PEAS ideas are from the reference below.